

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P2H Worked Solutions

Jan 2020 | P2H

33 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2H - Jan 2020 | P2H

1 (a) Simplify $\frac{x^9}{x^2}$

.....
(1)

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

.....
(2)

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

2. (b)

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

FINAL ANSWER(a) x^7 , (b) 7^9 .

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

2 Change 32.4 m^3 into cm^3 cm^3 **(Total for Question 2 is 2 marks)**

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$1 \text{ m} = 100 \text{ cm}$$

2. Calculate value

So

$$1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

$$32.4 \text{ m}^3 = 32.4 \times 1,000,000$$

$$= 32,400,000 \text{ cm}^3$$

FINAL ANSWER32,400,000 cm³.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Fractions**

Jan 2020 P2H - Jan 2020 | P2H

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Fractions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert to improper fractions**

Convert to improper fractions:

$$4\frac{2}{3} = \frac{14}{3}$$

$$3\frac{4}{5} = \frac{19}{5}$$

2. Use a common denominator

Use a common denominator:

$$\begin{aligned}\frac{14}{3} + \frac{19}{5} &= \frac{70}{15} + \frac{57}{15} \\ &= \frac{127}{15} = 8\frac{7}{15}\end{aligned}$$

FINAL ANSWER

$$8\frac{7}{15}$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jan 2020 P2H - Jan 2020 | P2H

- 4 The diagram shows a triangle.

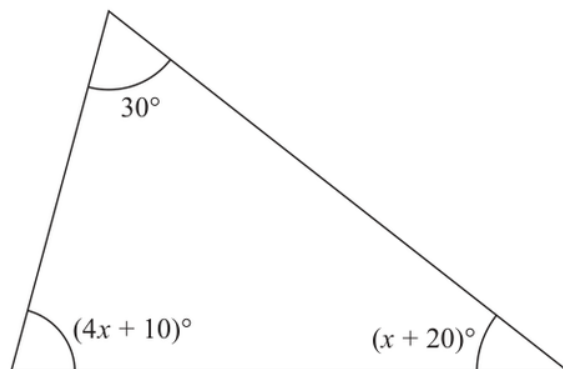


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Angles in a triangle addAngles in a triangle add to 180° :

$$30 + (4x + 10) + (x + 20) = 180$$

$$5x + 60 = 180$$

$$5x = 120$$

$$x = 24$$

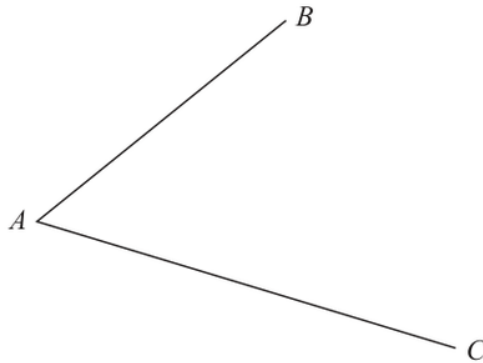
FINAL ANSWER

$$x = 24.$$

PAST PAPER QUESTION**Q#: 5****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

Jan 2020 P2H - Jan 2020 | P2H

- 5 Use ruler and compasses to construct the bisector of angle BAC .
You must show all your construction lines.



(Total for Question 5 is 2 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the bisector of angle BAC :

1. Put the compass point at A and draw an arc that cuts AB and AC .
2. From the two points where the arc cuts the arms, draw equal arcs inside the angle.
3. Draw a straight line from A through the intersection of those two arcs.

FINAL ANSWER

The ray from A through the arc intersection is the angle bisector of BAC .

EA

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Probability Toolkit**

Jan 2020 P2H - Jan 2020 | P2H

- 6** A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

Colour	red	blue	green	yellow
Probability		0.24		0.31

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.

She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Probability Toolkit**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the probability of red**

Let the probability of red be r . Then the probability of green is $2r$.

$$r + 0.24 + 2r + 0.31 = 1$$

$$3r = 0.45$$

$$r = 0.15$$

2. Expected number of red counters

Expected number of red counters:

$$180 \times 0.15 = 27$$

FINAL ANSWER

27.

PAST PAPER QUESTION**Q#: 7****Marks: 5****TOPIC: Solving Quadratic Equations**

Jan 2020 P2H - Jan 2020 | P2H

7 (a) Solve the inequality $2x + 7 > 4$

.....
(2)

(b) Solve $x^2 - 3x - 40 = 0$
Show clear algebraic working.

.....
(3)

(Total for Question 7 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 7****Marks: 5****TOPIC: Solving Quadratic Equations**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$2x + 7 > 4$$

$$2x > -3$$

$$x > -\frac{3}{2}$$

2. (b)

(b)

$$x^2 - 3x - 40 = 0$$

3. Factorise

Factorise:

$$(x - 8)(x + 5) = 0$$

4. Solve quadratic equation

So

$$x = 8 \quad \text{or} \quad x = -5$$

FINAL ANSWER

$$x > -\frac{3}{2}, \text{ and } x = 8 \text{ or } x = -5.$$

PAST PAPER QUESTION**Q#: 8****Marks: 7****TOPIC: Percentages**

Jan 2020 P2H - Jan 2020 | P2H

- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

"The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

- (a) Is Brigitte correct?

You must show how you get your answer.

(4)

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

- (b) Work out the discount that Henri gets.

..... euros

(3)

(Total for Question 8 is 7 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 7****TOPIC: Percentages**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

From 2016 to 2017, the increase is

$$545 - 500 = 45$$

$$\frac{45}{500} \times 100 = 9\%$$

2. Convert standard form

From 2017 to 2018, the increase is

$$592 - 545 = 47$$

$$\frac{47}{545} \times 100 \approx 8.62\%$$

3. Calculate percentage

So Brigitte is not correct because the second percentage increase is smaller.

4. Calculate percentage

For Henri, 952 is 85% of the original price because there is a 15% discount.

$$\text{original price} = 952 \div 0.85 = 1120$$

5. The discount is

The discount is

$$1120 - 952 = 168$$

FINAL ANSWER

Brigitte is not correct; the discount is 168 euros.

PAST PAPER QUESTION**Q#: 9****Marks: 2****TOPIC: Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

- 9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

..... g

(Total for Question 9 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 9 Marks: 2

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate volume

$$\text{mass} = \text{density} \times \text{volume}$$

$$\text{mass} = 19.3 \times 150$$

$$\text{mass} = 2895$$

FINAL ANSWER

2895 g.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

10 Change a speed of 50 metres per second to a speed in kilometres per hour.

..... kilometres per hour

(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert m/s to km/h multiply

To convert m/s to km/h, multiply by 3.6.

$$50 \times 3.6 = 180$$

FINAL ANSWER

180 km/h.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Area & Perimeter**

Jan 2020 P2H - Jan 2020 | P2H

- 11** The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

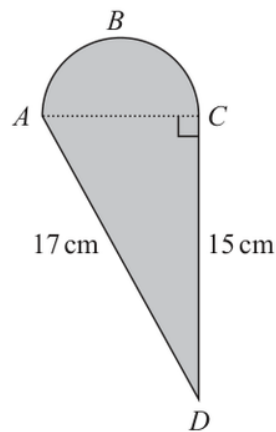


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Area & Perimeter**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**Triangle ACD is right-angled, with

$$AD = 17 \text{ cm}, \quad CD = 15 \text{ cm}$$

2. Use Pythagoras to findUse Pythagoras to find AC :

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

$$AC = 8 \text{ cm}$$

3. Calculate area

So the semicircle has radius

$$r = 4 \text{ cm}$$

4. The arc length of the

The arc length of the semicircle is

$$\pi r = 4\pi$$

5. The perimeter of the shaded

The perimeter of the shaded shape is

$$AD + CD + \text{semicircle arc} = 17 + 15 + 4\pi = 44.566\dots$$

FINAL ANSWER

44.6 cm.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Exchange Rates & Best Buys**

Jan 2020 P2H - Jan 2020 | P2H

- 12** Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

$$1 \text{ Dollar} = 6.57 \text{ Krone}$$

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 4

TOPIC: **Exchange Rates & Best Buys**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert standard form

Dane Oil costs 2500000 Krone for

$$4.2 \times 10^5 = 420000$$

2. litres

litres.

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

3. Convert standard form

So Dane Oil costs about 5.95 Krone per litre.

4. Arctic Oil costs dollars. Convert

Arctic Oil costs 770000 dollars. Convert this to Krone:

$$770000 \times 6.57 = 5058900$$

5. Arctic Oil gives

Arctic Oil gives

$$8.6 \times 10^5 = 860000$$

6. litres

litres.

$$\text{Arctic cost per litre} = \frac{5058900}{860000} = 5.882 \dots$$

7. Arctic Oil has the lower

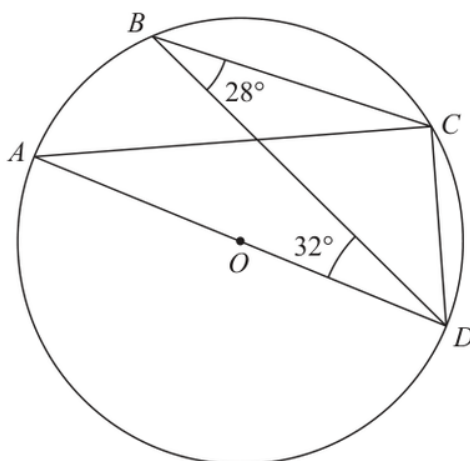
Arctic Oil has the lower cost per litre.

FINAL ANSWER

Arctic Oil

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Circle Theorems**

Jan 2020 P2H - Jan 2020 | P2H

13Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O .
 AOD is a diameter of the circle.

Angle $CBD = 28^\circ$

Angle $BDA = 32^\circ$

Find the size of angle BDC .

Give a reason for each stage of your working.

o

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 4****TOPIC: Circle Theorems**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**Since AD is a diameter,

$$\angle ABD = 90^\circ$$

2. Given

Given

$$\angle CBD = 28^\circ$$

3. Use trigonometry

So

$$\angle ABC = 90^\circ - 28^\circ = 62^\circ$$

4. Opposite angles in a cyclicOpposite angles in a cyclic quadrilateral add to 180° :

$$\angle ADC = 180^\circ - 62^\circ = 118^\circ$$

5. Use trigonometryAt D ,

$$\angle ADC = \angle BDA + \angle BDC$$

$$118^\circ = 32^\circ + \angle BDC$$

$$\angle BDC = 86^\circ$$

FINAL ANSWER 86° .

PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2H - Jan 2020 | P2H

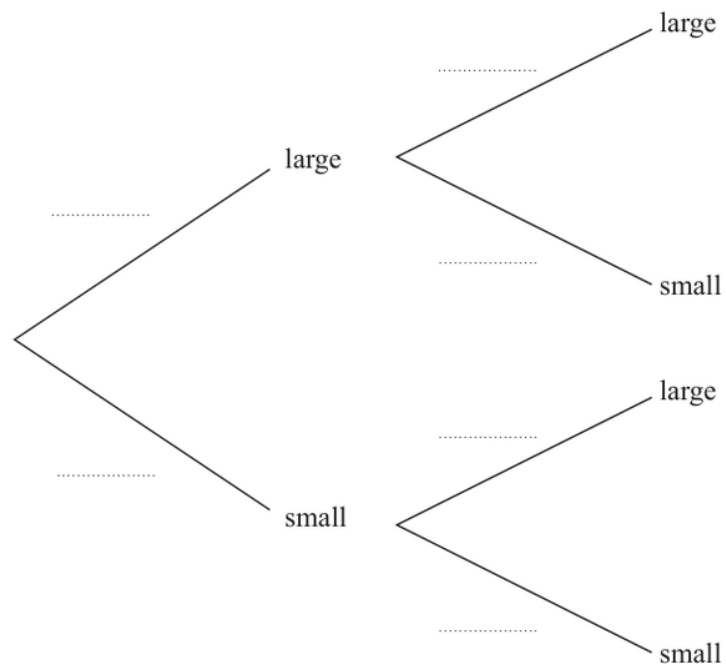
14 There are 20 glasses in a cupboard.

13 of the glasses are large

7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

(2)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

There are 13 large glasses and 7 small glasses.

$$\begin{aligned}P(\text{two small}) &= \frac{7}{20} \times \frac{6}{19} \\&= \frac{21}{190}\end{aligned}$$

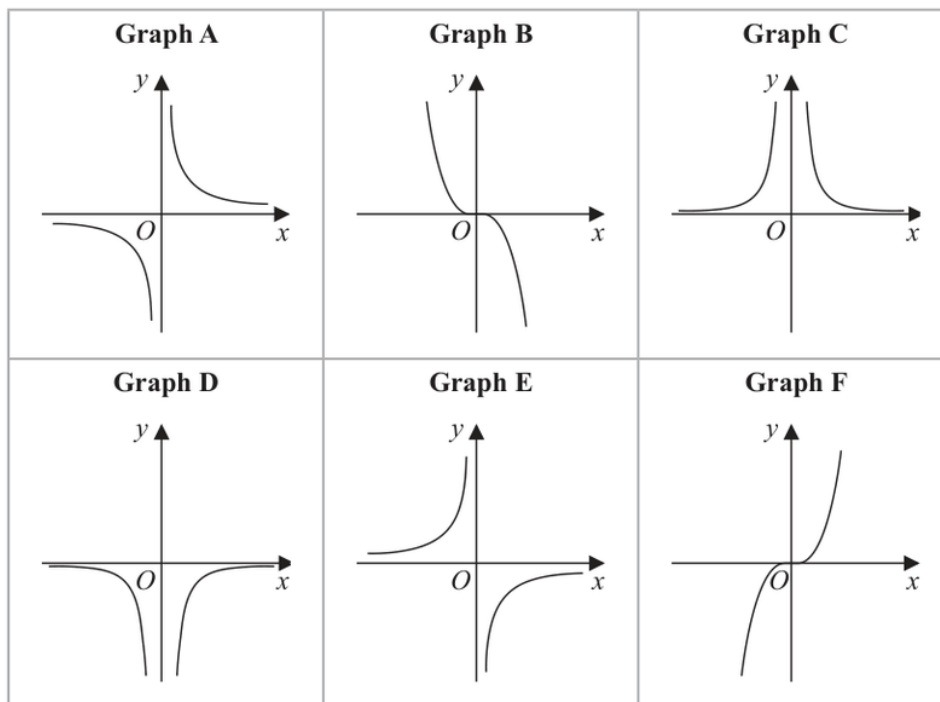
FINAL ANSWER

$$\frac{21}{190}$$

EA

PAST PAPER QUESTION**Q#: 15** **Marks: 3****TOPIC: Graphs of Functions**

Jan 2020 P2H - Jan 2020 | P2H

15 Here are six graphs.

Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	
$y = -\frac{1}{2}x^3$	
$y = -\frac{5}{x}$	

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Graphs of Functions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve equation**

For

$$y = -\frac{2}{x^2}$$

the graph is negative on both sides of the y -axis and is symmetrical in the y -axis.

2. This is Graph D

This is Graph D.

3. Solve equation

For

$$y = -\frac{1}{2}x^3$$

the graph is a decreasing cubic through the origin.

4. This is Graph B

This is Graph B.

5. Solve equation

For

$$y = \frac{5}{x}$$

the graph is a positive reciprocal graph in quadrants 1 and 3.

6. This is Graph A

This is Graph A.

FINAL ANSWER

D, B, A.

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Rearranging Formulas**

Jan 2020 P2H - Jan 2020 | P2H

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Rearranging Formulas**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Rearrange formula**

$$y = \sqrt{\frac{x+1}{x-4}}$$

2. Square both sides

Square both sides:

$$y^2 = \frac{x+1}{x-4}$$

$$y^2(x-4) = x+1$$

$$y^2x - 4y^2 = x+1$$

$$x(y^2 - 1) = 4y^2 + 1$$

FINAL ANSWER

$$x = \frac{4y^2+1}{y^2-1}$$

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

Jan 2020 P2H - Jan 2020 | P2H

- 17** Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17** **Marks: 3****TOPIC: Algebraic Proof**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the two consecutive square**

Let the two consecutive square numbers be n^2 and $(n + 1)^2$.

$$(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

2. is one more than an

$2n + 1$ is one more than an even number, so it is always odd.

FINAL ANSWER

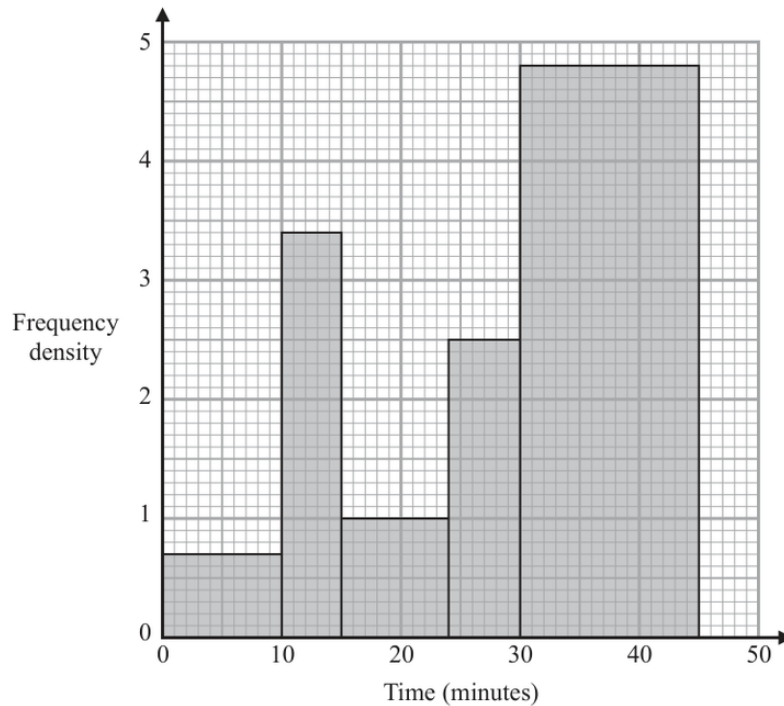
The difference is always odd.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Histograms**

Jan 2020 P2H - Jan 2020 | P2H

- 18** The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



- (a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

.....
(3)

One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

.....
(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Histograms**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Total frequency**

Total frequency:

$$10(0.7) + 5(3.4) + 9(1) + 6(2.5) + 15(4.8) = 120$$

2. Between 17 and 35 minutes

Between 17 and 35 minutes:

$$7(1) + 6(2.5) + 5(4.8) = 46$$

$$\text{proportion} = \frac{46}{120} = \frac{23}{60}$$

3. Given the customer spent more

Given the customer spent more than 30 minutes, they are in the 30 to 45 minute class.

4. More than 36 minutes is

More than 36 minutes is 9 minutes out of the 15-minute class:

$$\frac{9}{15} = \frac{3}{5}$$

FINAL ANSWER $\frac{23}{60}$, then $\frac{3}{5}$.

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

- 19** (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

.....
(1)

The line **L** passes through the points with coordinates $(-3, 1)$ and $(2, -2)$

- (b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates $(-6, 4)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

.....
(4)

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$y = 7 - 4x$$

2. A parallel line has theA parallel line has the same gradient, -4 . One possible answer is

$$y = -4x$$

3. (b) The gradient of line(b) The gradient of line L through $(-3, 1)$ and $(2, -2)$ is

$$\frac{-2 - 1}{2 - (-3)} = -\frac{3}{5}$$

4. A perpendicular line has gradientA perpendicular line has gradient $\frac{5}{3}$.**5. Evaluate fraction**Through $(-6, 4)$:

$$y - 4 = \frac{5}{3}(x + 6)$$

6. Multiply by 3

Multiply by 3:

$$3y - 12 = 5x + 30$$

$$5x - 3y + 42 = 0$$

FINAL ANSWERFor example $y = -4x$, and $5x - 3y + 42 = 0$.

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

20 The area of a rectangle is 18 cm^2 The length of the rectangle is $(\sqrt{7} + 1) \text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Surds**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

2. Rationalise the denominator

Rationalise the denominator:

$$\begin{aligned}\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} &= \frac{18(\sqrt{7} - 1)}{7 - 1} \\ &= \frac{18(\sqrt{7} - 1)}{6} = 3(\sqrt{7} - 1) \\ &= 3\sqrt{7} - 3\end{aligned}$$

FINAL ANSWER

$$3\sqrt{7} - 3 \text{ cm}$$

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

20 The area of a rectangle is 18 cm^2 The length of the rectangle is $(\sqrt{7} + 1) \text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20** **Marks: 3****TOPIC: Surds**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Area of rectangle**

Area of rectangle:

$$18 \text{ cm}^2$$

2. Length

Length:

$$\sqrt{7} + 1$$

3. Width

Width:

$$\frac{18}{\sqrt{7} + 1}$$

4. Rationalise the denominator

Rationalise the denominator:

$$\begin{aligned} & \frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} \\ &= \frac{18(\sqrt{7} - 1)}{7 - 1} \\ &= \frac{18(\sqrt{7} - 1)}{6} \\ &= 3(\sqrt{7} - 1) \\ &= 3\sqrt{7} - 3 \end{aligned}$$

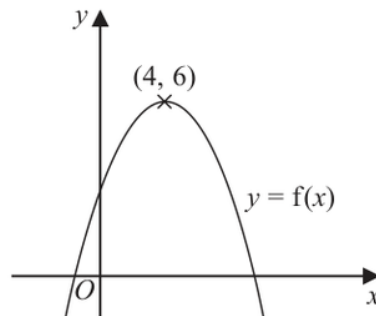
FINAL ANSWER

$$3\sqrt{7} - 3 \text{ cm}$$

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

- 21** The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

- (a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)
(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

- (b) Find an equation of curve **S**.

You do not need to simplify the equation.

.....
(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(4, 6)$.**2. Calculate value**

For

$$y = f(x + 4)$$

3. the graph moves 4 units

the graph moves 4 units left:

$$(4, 6) \rightarrow (0, 6)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 6) \rightarrow (2, 6)$$

6. Calculate value

For part (b),

$$y = x^2 + 3x + 4$$

7. The translation vector is

The translation vector is

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

8. Read the graph

So the graph moves 4 units right and 6 units up.

9. Add 6Replace x by $x - 4$, then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

FINAL ANSWER

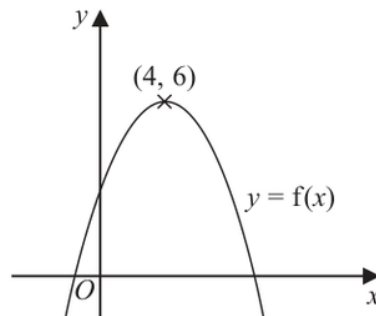
$(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

- 21** The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

- (a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)
(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

- (b) Find an equation of curve **S**.

You do not need to simplify the equation.

.....
(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(4, 6)$.**2. Calculate value**

For

$$y = f(x + 4)$$

3. the graph moves 4 units

the graph moves 4 units left:

$$(4, 6) \rightarrow (0, 6)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 6) \rightarrow (2, 6)$$

6. Calculate value

For part (b),

$$y = x^2 + 3x + 4$$

7. The translation vector is

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$$y = (x - 4)^2 + 3(x - 4) + 10$$

FINAL ANSWER

$(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

- 22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .
Show clear algebraic working.

(.....,)

(.....,)

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The line is**

The line is

$$y = x + 2$$

2. Substitute this into the circle

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x + 2)^2 - 2(x + 2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0$$

3. Calculate value

So

$$x = -4 \quad \text{or} \quad x = 3$$

4. Calculate valueUsing $y = x + 2$:

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

FINAL ANSWER $(-4, -2)$ and $(3, 5)$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

- 22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .
Show clear algebraic working.

(.....,)

(.....,)

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The line is**

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$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

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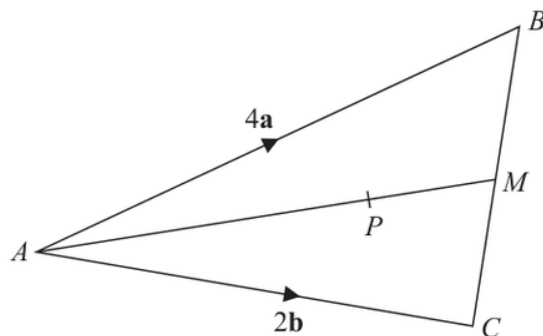
$$x = -4 \Rightarrow y = -2$$

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FINAL ANSWER $(-4, -2)$ and $(3, 5)$

PAST PAPER QUESTION**Q#: 23** **Marks: 3**TOPIC: **Vectors**

Jan 2020 P2H - Jan 2020 | P2H

23Diagram **NOT**
accurately drawn

ABC is a triangle.
The midpoint of BC is M .
 P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Vectors**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**Since M is the midpoint of BC ,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

2. Given

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

3. Factorise

Factorise:

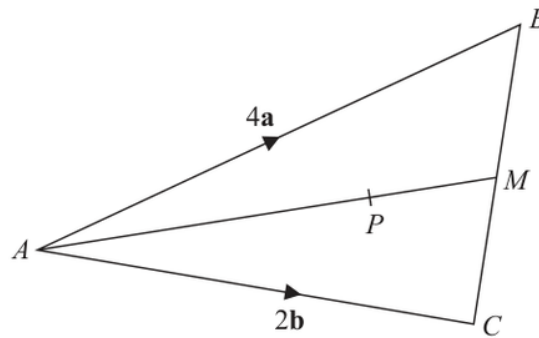
$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

4. Split the ratioSo AP is $\frac{3}{4}$ of AM , leaving PM as $\frac{1}{4}$ of AM .**FINAL ANSWER**

$$AP : PM = 3 : 1$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Vectors**

Jan 2020 P2H - Jan 2020 | P2H

23Diagram **NOT**
accurately drawn

ABC is a triangle.
 The midpoint of BC is M .
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$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Vectors**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**Since M is the midpoint of BC ,

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Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

4. Split the ratioSo AP is $\frac{3}{4}$ of AM , leaving PM as $\frac{1}{4}$ of AM .**FINAL ANSWER**

$$AP : PM = 3 : 1$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

24 Express

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

2. Use trigonometry

First simplify the bracket:

$$\begin{aligned} \frac{4}{2x-5} - \frac{3}{2x-3} &= \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)} \\ &= \frac{8x-12-6x+15}{(2x-5)(2x-3)} \\ &= \frac{2x+3}{(2x-5)(2x-3)} \end{aligned}$$

3. Use trigonometry

Now factor the second fraction:

$$9x-4x^3 = -x(2x-3)(2x+3)$$

$$6x^2-17x+5 = (2x-5)(3x-1)$$

4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)} \times \frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

5. Cancel common factors

Cancel common factors:

$$\begin{aligned} &= -\frac{3x-1}{x(2x-3)^2} \\ &= \frac{1-3x}{x(2x-3)^2} \end{aligned}$$

FINAL ANSWER

$$\frac{1-3x}{x(2x-3)^2}$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

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(Total for Question 24 is 4 marks)

EA

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Jan 2020 P2H - Jan 2020 | P2H

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FINAL ANSWER

$$\frac{1-3x}{x(2x-3)^2}$$

PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Sequences**

Jan 2020 P2H - Jan 2020 | P2H

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$k = \dots\dots\dots$

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Sequences**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Mario saves from 2021 up**

Mario saves from 2021 up to and including 2070.

2. Number of years

Number of years:

$$2070 - 2021 + 1 = 50$$

3. This is an arithmetic sequence

This is an arithmetic sequence.

4. Use trigonometry

First term:

$$50$$

5. Common difference

Common difference:

$$k$$

6. Last term

Last term:

$$50 + 49k$$

7. Use the arithmetic series formula

Use the arithmetic series formula:

$$S_n = \frac{n}{2}(a + l)$$

$$33125 = \frac{50}{2} (50 + (50 + 49k))$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

FINAL ANSWER

$$k = 25$$

EA

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: Sequences

Jan 2020 P2H - Jan 2020 | P2H

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Work out the value of k .

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(Total for Question 25 is 3 marks)

EA

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Jan 2020 P2H - Jan 2020 | P2H

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$$33125 = 25(100 + 49k)$$

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$$49k = 1225$$

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FINAL ANSWER

$$k = 25$$

EA

PAST PAPER QUESTION**Q#: 26****Marks: 6****TOPIC: Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

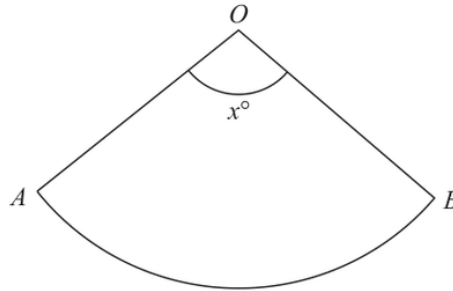


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

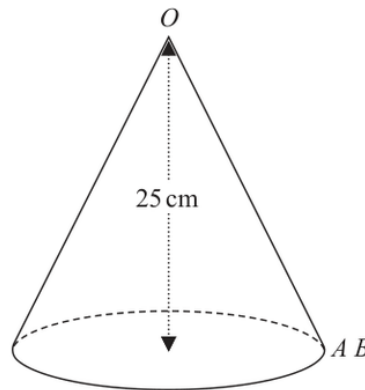


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm.

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

(Total for Question 26 is 6 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 6****TOPIC: Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the radius of the**

Let the radius of the base of the cone be r , and let the slant height be l .

2. The volume of the cone

The volume of the cone is

$$\frac{1}{3}\pi r^2 h = 1600$$

3. Calculate area

The height is 25 cm, so

$$\frac{1}{3}\pi r^2 (25) = 1600$$

$$r^2 = \frac{4800}{25\pi}$$

$$r = 7.8176\dots$$

4. The slant height is found

The slant height is found using Pythagoras:

$$l^2 = r^2 + 25^2$$

$$l = 26.1938\dots$$

5. Calculate area

When the sector is joined to form the cone, the arc length of the sector becomes the circumference of the cone base.

$$\frac{x}{360} \times 2\pi l = 2\pi r$$

6. Cancel

Cancel 2π :

$$\frac{x}{360} l = r$$

$$x = \frac{360r}{l}$$

$$x = \frac{360(7.8176\dots)}{26.1938\dots} = 107.4433\dots$$

FINAL ANSWER

107°

EA

PAST PAPER QUESTION**Q#: 26** **Marks: 6****TOPIC: Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

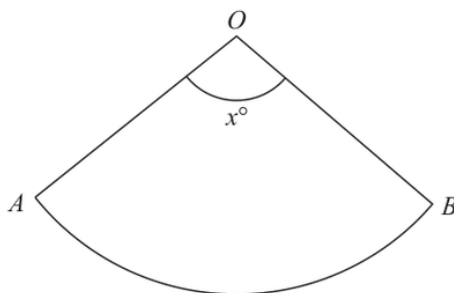


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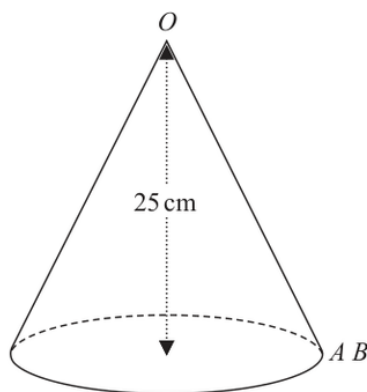


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(Total for Question 26 is 6 marks)

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Jan 2020 P2H - Jan 2020 | P2H

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$$x = \frac{360(7.8176\dots)}{26.1938\dots} = 107.4433\dots$$

FINAL ANSWER

107°

EA